

Production RAG Specification

Retrieval Architecture

The production retrieval pipeline uses query embeddings and PostgreSQL pgvector to locate relevant document chunks. A chat request is authenticated before document ownership is resolved. The retriever receives the authenticated user ID, document ID, query vector, and requested top-K value. Retrieval is scoped to the requested document and authenticated user.

Dense Retrieval

Dense retrieval compares the query embedding with stored chunk embeddings using cosine distance. Candidates are ordered by ascending distance. The production retriever applies a best distance threshold of 0.70 and an adaptive margin of 0.15. The adaptive limit is the smaller of the best observed distance plus the margin and the maximum allowed distance.

Candidate Limits

The retriever accepts a top-K value from 1 through 50. The application normally requests a small candidate set because retrieval quality must be balanced against context size and downstream processing cost. The final result count can be lower than the requested top-K value when adaptive distance filtering removes weak matches.

Document Isolation

Document retrieval must preserve the authenticated ownership boundary. A document ID is not an authorization credential. Candidate queries must constrain the document relationship and the owning user relationship. Retrieval must not obtain broad cross-tenant candidates and filter them only after retrieval.

Context Construction

Retrieved chunks are converted into context for the language model. Chunk ordering and document identity remain available during context construction. The retrieval layer is responsible for returning only chunks that satisfy its ownership and distance constraints.

Evaluation Metrics

Retrieval quality is evaluated using Hit Rate at K, Mean Reciprocal Rank, Context Precision at K, and Context Recall at K. Primary evaluation uses K equal to 5, while K values of 1, 3, 5, and 10 provide ranking and recall characterization. Golden queries identify one or more relevant chunk IDs.

Failure Modes

A retrieval failure can occur when the relevant chunk is absent from the returned candidate set, ranked too low, or rejected by the configured distance threshold. Evaluation should distinguish retrieval misses from downstream generation failures. A low retrieval score should not automatically be interpreted as an LLM quality problem.

Hybrid Retrieval Evaluation

Hybrid retrieval combines lexical and dense signals to test whether exact terminology and semantic similarity provide complementary evidence. Lexical retrieval is particularly useful for identifiers, version strings, error codes, and exact terms, while dense retrieval can improve meaning-level matching.

Reranking Evaluation

A reranker may reorder an initial candidate set using richer query-document relevance features. Evaluation must compare dense retrieval, lexical retrieval, hybrid retrieval, dense retrieval with reranking, and hybrid retrieval with reranking using the same golden query set. Reranking must not weaken document ownership or tenant-isolation guarantees.

Production Decision Criteria

A retrieval architecture should be selected using quality, latency, reliability, and security evidence rather than a single metric. Improvements in recall are not sufficient if latency or isolation guarantees regress. The evaluation framework therefore records ranking metrics, failure modes, and retrieval latency before recommending a production configuration.